

Geometric Constructions Using Triangle Congruence

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Geometric construction is the process of drawing figures using only a compass and a straightedge.

Constructing Segments

A **segment** can be copied or bisected accurately using geometric construction.

$$\overline{AB} = 8 \text{ cm}$$

A congruent segment has the same length.

Constructing Angles

An **angle** can be copied or bisected using a compass and straightedge.

$$\angle XYZ = 58^\circ$$

A congruent angle has the same measure.

Constructing Triangles

A **triangle** can be constructed when sufficient measurements are given.

Common conditions include

SSS

SAS

ASA

Triangle Congruence

The **triangle congruence postulates and theorems** justify why a constructed triangle is unique.

If the required corresponding parts are congruent, the triangles are congruent.

Secondary Parts

The **perpendicular bisectors**, **angle bisectors**, **medians**, and **altitudes** are called the **secondary parts** of a triangle.

Triangle Centers

The secondary parts intersect at special points called the **triangle centers**.

Circumcenter

Intersection of perpendicular bisectors.

Triangle Centers

Incenter

Intersection of angle bisectors.

Centroid

Intersection of medians.

Triangle Centers

Orthocenter

Intersection of altitudes.

Examples

Example 1

Construct a segment congruent to

$$\overline{AB} = 8 \text{ cm.}$$

The constructed segment must have the same length.

8 cm

Example 2

Construct the perpendicular bisector of a segment measuring 10 cm.

The perpendicular bisector divides the segment into two congruent parts.

$$\frac{10}{2} = 5$$

5 cm each

Example 3

Construct an angle congruent to

$$\angle XYZ = 58^\circ.$$

The constructed angle has the same measure.

$$\boxed{58^\circ}$$

Example 4

Construct the bisector of a 96° angle.

The angle is divided into two congruent angles.

$$\frac{96^\circ}{2} = 48^\circ$$

48° each

Example 5

Construct a triangle with side lengths 5 cm, 6 cm, and 8 cm.

Three sides are given, so the congruence postulate is

SSS

SSS guarantees a unique triangle.

Example 6

Construct a triangle with two sides measuring 7 cm and 9 cm and the included angle measuring 55° .

Two sides and the included angle are given.

SAS

Practice Exercises

Practice 1

Construct a segment congruent to

$$\overline{AB} = 8 \text{ cm.}$$

Practice 2

Construct the perpendicular bisector of a segment measuring 10 cm.

Determine the length of each resulting segment.

Practice 3

Construct an angle congruent to

$$\angle XYZ = 58^\circ.$$

Practice 4

Construct the bisector of a 96° angle.
Find the measure of each resulting angle.

Practice 5

Construct a triangle with side lengths 5 cm, 6 cm, and 8 cm.

Identify the congruence postulate that guarantees a unique triangle.

Practice 6

Construct a triangle with two sides measuring 7 cm and 9 cm and the included angle measuring 55° .

Activities

Activity 1

Construct a segment congruent to

$$\overline{CD} = 7 \text{ cm.}$$

Activity 2

Construct the perpendicular bisector of a segment measuring 14 cm.

Determine the length of each resulting segment.

Activity 3

Construct an angle congruent to

$$\angle ABC = 72^\circ.$$

Activity 4

Construct the bisector of a 124° angle.
Find the measure of each resulting angle.

Activity 5

Construct a triangle with side lengths 4 cm, 7 cm, and 9 cm.

Identify the congruence postulate that guarantees a unique triangle.

Activity 6

Construct a triangle with two sides measuring 8 cm and 10 cm and the included angle measuring 68° .

Thank you!